

1 Linear Combinations

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |0220| \cdot |aaaa| + 1 \cdot |aaaa| \cdot |0220|$$

or, expressed in terms of symmetry: monomer determinants:

$$+ \quad 1 \quad \cdot \quad |(b_{1u}^r) (b_{2g}^r) (b_{3g}^r) (a_u^r)| \quad + \quad 1 \quad \cdot \quad |(b_{1u}^l) (b_{2g}^l) (b_{3g}^l) (a_u^l)|$$

substitution by dimer orbitals:

$$+1 \cdot |(-a_g + b_{1u}) (-b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned} & + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ & + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\ & + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\ & + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\ & + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\ & \quad + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{aligned}$$

summarized:

$$\begin{aligned} & +2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\ & \quad + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \end{aligned}$$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |0220| \cdot |aaaa| - 1 \cdot |aaaa| \cdot |0220|$$

or, expressed in terms of symmetry: monomer determinants:

$$+ 1 \cdot |(b_{1u}^r)(b_{2g}^r)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^l)(a_u^l)|$$

substitution by dimer orbitals:

$$+1 \cdot |(-a_g + b_{1u})(-b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(+b_{2u} + b_{3g})(+b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned} & + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \\ & + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \\ & + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\ & - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \\ & - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \\ & - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \end{aligned}$$

summarized:

$$\begin{aligned} & -2 \cdot \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - 2 \cdot \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - 2 \cdot \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \\ & - 2 \cdot \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - 2 \cdot \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \end{aligned}$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |0a2a|$$

or, expressed in terms of symmetry: monomer determinants:

$$+1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|$$

substitution by dimer orbitals:

$$+1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0\rangle \cdot |a2a0\rangle + 1 \cdot |a2a0\rangle \cdot |a2a0\rangle + 1 \cdot |a2a0\rangle \cdot |0a2a\rangle + 1 \cdot |0a2a\rangle \cdot |a2a0\rangle \\
& -1 \cdot |0a2a\rangle \cdot |a2a0\rangle - 1 \cdot |a2a0\rangle \cdot |0a2a\rangle - 1 \cdot |0a2a\rangle \cdot |0a2a\rangle - 1 \cdot |0a2a\rangle \cdot |0a2a\rangle
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| + 1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)| - 1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

[illegible]

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |a2a0| \\
& -1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| - 1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)| + 1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

[illegible]

summarized:

$$\begin{aligned}
& -4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& - 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0| \\
& -1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& - \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \\
& + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \\
& + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}
\end{aligned}$$

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\
& +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& +2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0| \\
& -1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} + \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \\
& + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \\
& + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}
\end{aligned}$$

summarized:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0| \\
& -1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& + \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0| \\
& - 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& - 1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& - 1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} + \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \\
& + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \\
& + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}
\end{aligned}$$

summarized:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right)$$

written in CI vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |0a2a|$$

or, expressed in terms of symmetry: monomer determinants:

$$+1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|$$

substitution by dimer orbitals:

$$+1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{2u} b_{2u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3g} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2g} b_{2g} a_u a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0\rangle \cdot |aa20\rangle + 1 \cdot |aa20\rangle \cdot |a2a0\rangle - 1 \cdot |a2a0\rangle \cdot |02aa\rangle - 1 \cdot |02aa\rangle \cdot |a2a0\rangle \\
& +1 \cdot |0a2a\rangle \cdot |aa20\rangle + 1 \cdot |aa20\rangle \cdot |0a2a\rangle - 1 \cdot |0a2a\rangle \cdot |02aa\rangle - 1 \cdot |02aa\rangle \cdot |0a2a\rangle
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& - \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\
& +2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|
\end{aligned}$$

$$\begin{aligned}
& e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0| \\
& +1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| - 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| - 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} + \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \\
& - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}
\end{aligned}$$

summarized:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|
\end{aligned}$$

$$\begin{aligned}
& e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |02aa| \cdot |a2a0| \\
& +1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| + 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \\
& - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \\
& - \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \\
& + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \\
& - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}
\end{aligned}$$

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}} \right|
\end{aligned}$$

$$\begin{aligned}
& e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow & \uparrow & \uparrow & \downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow & \downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow & \uparrow & \downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow & \uparrow & \downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow & \uparrow & \downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow & \uparrow & \downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow & \downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow & \uparrow & \uparrow & \downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0| \\
& +1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{3g}^r)| + 1 \cdot |(b_{1u}^l)(b_{3g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{2g}^r)(a_u^r)| + 1 \cdot |(b_{2g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{2g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} - \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}} + \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} + \underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| \\
& + \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} + \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& - \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} + \underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} - \underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& -2 \cdot \left| \underbrace{a_g a_g b_{2u} b_{2g}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3g} b_{3g} a_u}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3u} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2g} b_{2g} a_u}_{a_u} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{3g} a_u a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2g} a_u a_u}_{a_u} \right|
\end{aligned}$$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |02aa| \cdot |02aa|$$

or, expressed in terms of symmetry: monomer determinants:

$$+1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)|$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| \\
& - 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& + \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |aa20| \\
& -1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| \\
& -1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| \\
& +1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| \\
& -1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

[illegible]

summarized:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g a_g b_{3u} b_{3u}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g a_g b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{2u} a_u a_u}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3g} b_{3g} a_u a_u}_{a_g} \right|
\end{aligned}$$

$$\begin{aligned}
& e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$\begin{aligned}
& +1 \cdot |aa20| \cdot |aa20| - 1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |aa20| \\
& -1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|
\end{aligned}$$

or, expressed in terms of symmetry: monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| - 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| \\
& -1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| - 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)|
\end{aligned}$$

substitution by dimer orbitals:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| \\
& +1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| \\
& -1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| \\
& -1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

[illegible]

summarized:

$$\begin{aligned}
& -4 \cdot \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| \\
& - 4 \cdot \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right|
\end{aligned}$$

$$\begin{aligned}
& i^3 b_{3u} * i^3 b_{3u} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |02aa| \cdot |02aa|$$

or, expressed in terms of symmetry: monomer determinants:

$$+1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{1u}^l)(b_{2g}^l)(b_{3g}^r)(a_u^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{1u}^r)(b_{2g}^r)| + 1 \cdot |(b_{3g}^l)(a_u^l)(b_{3g}^r)(a_u^r)|$$

substitution by dimer orbitals:

$$+1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u})(b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(b_{2u} + b_{3g})(b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

multiplied out:

